William Olson

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Education

University of Texas at San Antonio *Sept 2024 – May 2028*
Bachelor of Science in Physics – Minors in Mathematics and Astrophysics *Expected graduation*

- Cumulative GPA: **4.00/4.00**
- Relevant Coursework: Classical Mechanics, Mathematical Physics, Differential Equations, Linear Algebra

Honors & Awards

Distinguished Presidential Scholarship *2024 – Present*
- Competitive merit scholarship awarded by the University of Texas at San Antonio.

Texas Excellence Grant *2024 – Present*
- Financial support for undergraduate study.

University Honors College *2024 – Present*
- Selected member of the UTSA Honors College.

NSF REU Participant – Breakthrough Listen Initiative *Summer 2026*
- Selected to conduct summer research with the Breakthrough Listen Initiative at the University of California, Berkeley.

Research Experience

NSF REU Student Researcher – Breakthrough Listen Initiative *Berkeley, CA*
June – Aug 2026

- Conducted radio technosignature searches using Green Bank Telescope observations as part of the Breakthrough Listen collaboration.
- Evaluated narrowband search workflows using turboSETI and BLISS, comparing high- and mid-frequency-resolution data products with known spacecraft transmissions.
- Tested signal recovery and candidate validation using Voyager observations, ON/OFF comparisons, and synthetic signal injections.

Machine Learning for AGN Ionization Cone Detection *San Antonio, TX*
Sept 2024 – Present

- Develop U-Net deep-learning models for automated identification and segmentation of ionization cones in active galactic nuclei.
- Design a modular Python framework for synthetic JWST image generation, augmentation, model training, evaluation, and inference.
- Apply trained models to JWST NIRCam and MIRI observations to investigate generalization between simulated and observational datasets.
- Proposed using CLOUDY to model emission from regions with different physical conditions and improve the realism of synthetic training images.

AGN Torus and Polar Wind Modeling *San Antonio, TX*
Sept 2024 – Present

- Model infrared emission from AGN dust structures using `FortesFit` with CAT3D, CAT3D-Wind, and CLUMPY radiative transfer models.
- Perform forward modeling and image deconvolution to investigate dust geometry and polar emission.
- Compare theoretical model predictions with JWST observations to support interpretation of machine-learning analyses.

Technical Skills

Programming Languages: Python, Bash

Scientific Computing: NumPy, SciPy, Pandas, Astropy, GNU Radio

Machine Learning: PyTorch, U-Net, `segmentation_models_pytorch`

Astronomy Software & Data: MAST Archive, SAOImage DS9, FITS, GNU Astronomy Utilities (NoiseChisel)

Radio Astronomy: turboSETI, BLISS, blimpy, filterbank analysis, spectral analysis

Visualization: Matplotlib

Development Tools: Git, Linux, remote SSH workflows, Jupyter Notebook, LaTeX

Research Interests

Active Galactic Nuclei (AGN); galaxy evolution; astronomical machine learning; scientific artificial intelligence; radio astronomy; technosignature searches; JWST data analysis; astronomical instrumentation

Open-Source Software & Programs

Ionization Cone Automation Pipeline GitHub repository

- Developed a modular Python framework for synthetic AGN image generation, machine-learning training, inference, and evaluation.
- Implemented reusable workflows for astronomical image processing, data augmentation, and performance assessment.
- Designed to support reproducible deep-learning experiments on JWST observations.

Honors College Event Analyzer GitHub repository

- Developed a Python application that analyzes attendance trends across Honors College events.
- Performs statistical analysis using event categories, SPICES themes, and keyword relationships.
- Generates visualizations to support data-driven planning and assessment of student engagement.

Radio Signal Detection with NoiseChisel

- Developed a workflow to convert radio time–frequency data to FITS images and identify candidate signal structures using NoiseChisel.
- Applied robust row-based normalization and preprocessing to reduce the influence of instrumental artifacts on detection.

Presentations

Olson, W., Schneider, J., Zhang, L., Packham, C.
A Physically Motivated Machine Learning Framework for Systematic Identification of AGN Ionization Cones.
Poster presented at the Kyoto Torus Workshop, Kyoto, Japan, May 2026. **(International)**

Ortuno Lopez, L., **Olson, W.**, Schneider, J., Zhang, L., Gutierrez, J. B., Packham, C.
Using Artificial Intelligence to Leverage JWST Datasets to Explore Active Galactic Nuclei.
Poster presented at the Cosmic Horizons Conference, Austin, Texas, May 2025. **(National)**

Olson, W., Schneider, J., Zhang, L., Packham, C.
Physics-Guided Machine Learning for Automated Tracing of AGN Ionization Cones.
Poster presented at the UTSA–SwRI Symposium, San Antonio, Texas, April 2026. **(Community)**

Olson, W., Schneider, J., Zhang, L., Packham, C.
A Physically Motivated Machine Learning Framework for Systematic Identification of AGN Ionization Cones.
Poster presented at the UTSA Spring Symposium, San Antonio, Texas, April 2026. **(University)**

Olson, W., Schneider, J., et al.
LIRA: Lunar Infrared Array Mission Concept.
Oral presentation for the UTSA Leadership and Science Integration Course, May 2025. **(University)**

Olson, W., Jones, M.
Water Quality Assessment of Lake Boerne.

Poster presented at the APPEX Student Research Symposium, San Antonio, Texas, October 2024. **(University)**

Programs & Training

Academic Personal and Professional Excellence (APPEX) *Aug 2024 – Present*

- Selected for a competitive Ph.D. preparation program supporting undergraduate researchers pursuing graduate education.
- Completed training in scientific communication, research ethics, graduate fellowship preparation, and professional development.
- Participated in workshops on NSF Graduate Research Fellowship Program (GRFP) applications, graduate school preparation, and research presentation skills.
- Eligible for competitive conference travel funding through the APPEX program.

College of Science Work Study Research Training Program (COS WS-RTP) *Jan 2026 – Present*

- Selected for a competitive undergraduate research training program emphasizing laboratory readiness and professional development.
- Completed training in responsible conduct of research, laboratory safety, scientific communication, and graduate school preparation.
- Participated in seminars connecting undergraduate researchers with faculty mentors and research opportunities across the College of Science.

Invited Speaking & Outreach

Invited Student Speaker — College of Sciences Advancement Council *Sept 30, 2026*

University of Texas at San Antonio; Club Giraud, San Antonio, TX

- Selected to address donors, alumni, community leaders, and college leadership about undergraduate research and student success.
- Discussed how APPEX and the College of Sciences Student Success Center's Work Study Research Program supported early research involvement and academic and professional growth.

Additional Outreach & Mentoring

- Deliver STEM outreach talks at James Madison High School about college and pathways to a Ph.D.
- Volunteer track and field coach, mentoring student athletes in sprint training, technique development, and athletic preparation.

Affiliations

- UTSA Mid-Infrared Astrophysics & Exoplanet Science (MAE) Research Group – Student Researcher
- Breakthrough Listen Collaboration – NSF REU Participant
- Society of Physics Students (SPS), University of Texas at San Antonio
- Academic Personal and Professional Excellence (APPEX) Ph.D. Training Program
- College of Science Work Study Research Training Program (WSRTP)
- Honors College Student Advisory Board (HCSAB) – Past Member